

Term Spread

U.S. GDP

```
library(xts)
```

```
df <- readxl::read_excel("week 11 Term Spread.xlsx")
```

```
# Rename first column  
names(df)[1] <- "Quarter"
```

```
# Convert quarterly labels such as 1960:01 into year-quarter format  
year <- as.numeric(substr(df$Quarter, 1, 4))  
quarter <- as.numeric(substr(df$Quarter, 6, 7))
```

```
qtr_index <- as.yearqtr(year + (quarter - 1) / 4)
```

```
# Create xts object with quarterly index
```

```
macro.xts <- xts(  
  as.data.frame(df[, c("GS10", "TB3MS", "RSPREAD", "RECESSION", "GDP")]),  
  order.by = qtr_index  
)
```

> tail(macro.xts)

LR-SR
↓

		GS10	TB3MS	RSPREAD	RECESSION	GDP
2016	Q3	1.563333	0.2966667	1.266667 > 0	0	16778.15
2016	Q4	2.130000	0.4300000	1.700000 > 0	0	16851.42
2017	Q1	2.443333	0.5900000	1.853333 > 0	0	16903.24
2017	Q2	2.263333	0.8900000	1.373333 > 0	0	17031.08
2017	Q3	2.243333	1.0366667	1.206667 > 0	0	17163.89
2017	Q4	2.370000	1.2066667	1.163333 > 0	0	17271.70

GS10 is 10-Year Interest Rate LR

TB3MS is 3-Month Interest Rate SR

Rspread is GS10-TB3MS

Recession is an indicator

GDP is U.S. GDP (billions of \$2009) - real GDP

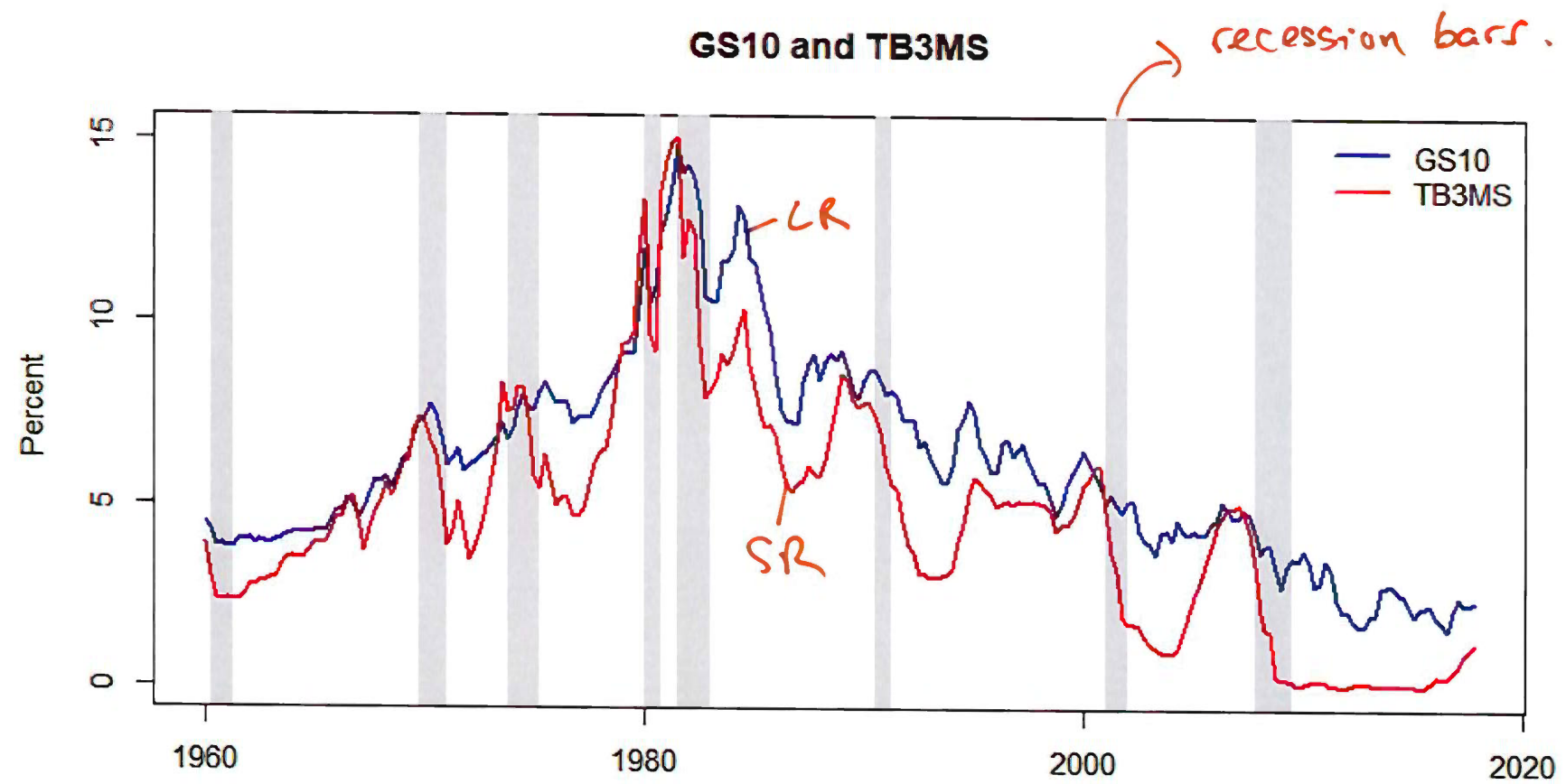
usually
LR > SR

Forecasting GDP Growth Using the Term Spread

Interest rates on long-term and short-term bonds move together but not one for one.

The figure plots interest rates on 10-year U.S. Treasury bonds and 3-month Treasury bills from 1960 through 2017.

These interest rates show the same long-run tendencies: Both were low in the 1960s, both rose through the 1970s and peaked in the early 1980s, and both fell subsequently.



Term Spread

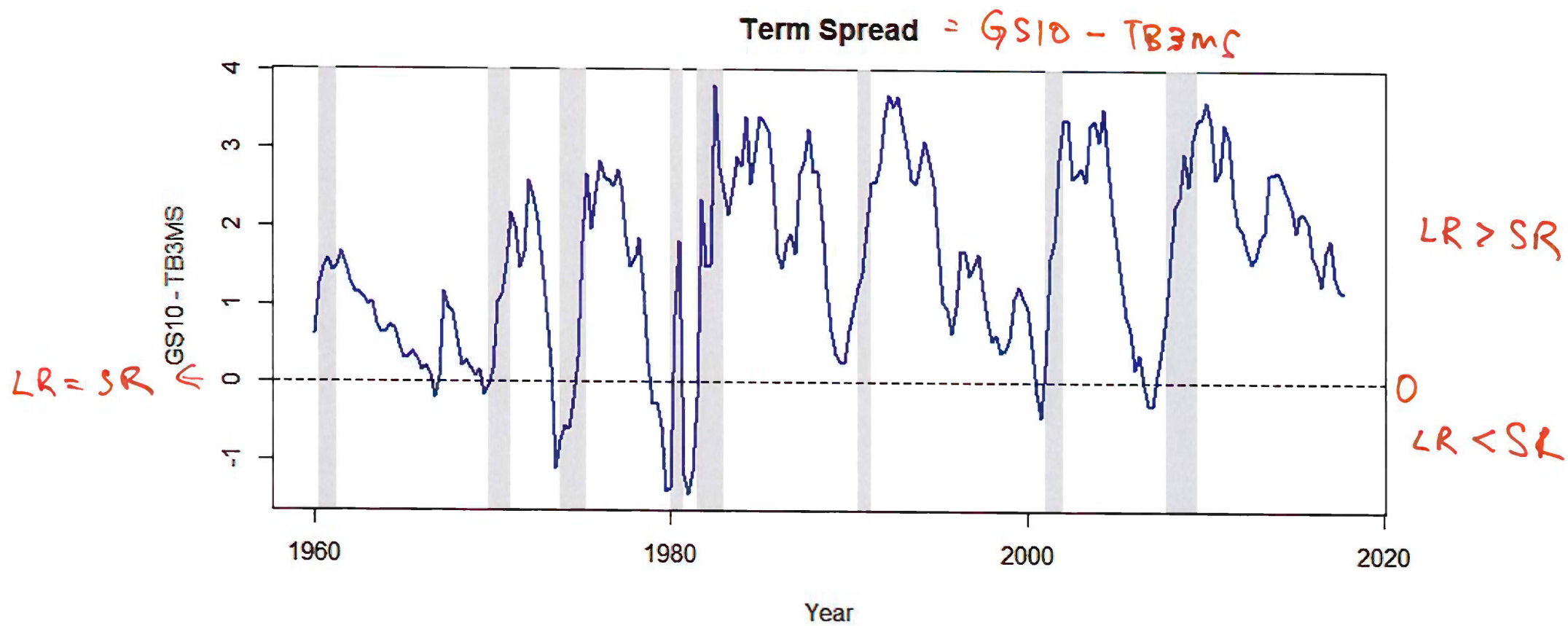
But the gap, or difference, between the two interest rates has not been constant:

While short-term rates are generally below long-term rates, the gap between them narrows and even disappears shortly before the start of a recession.

Recessions are shown as the shaded bars in the figure.

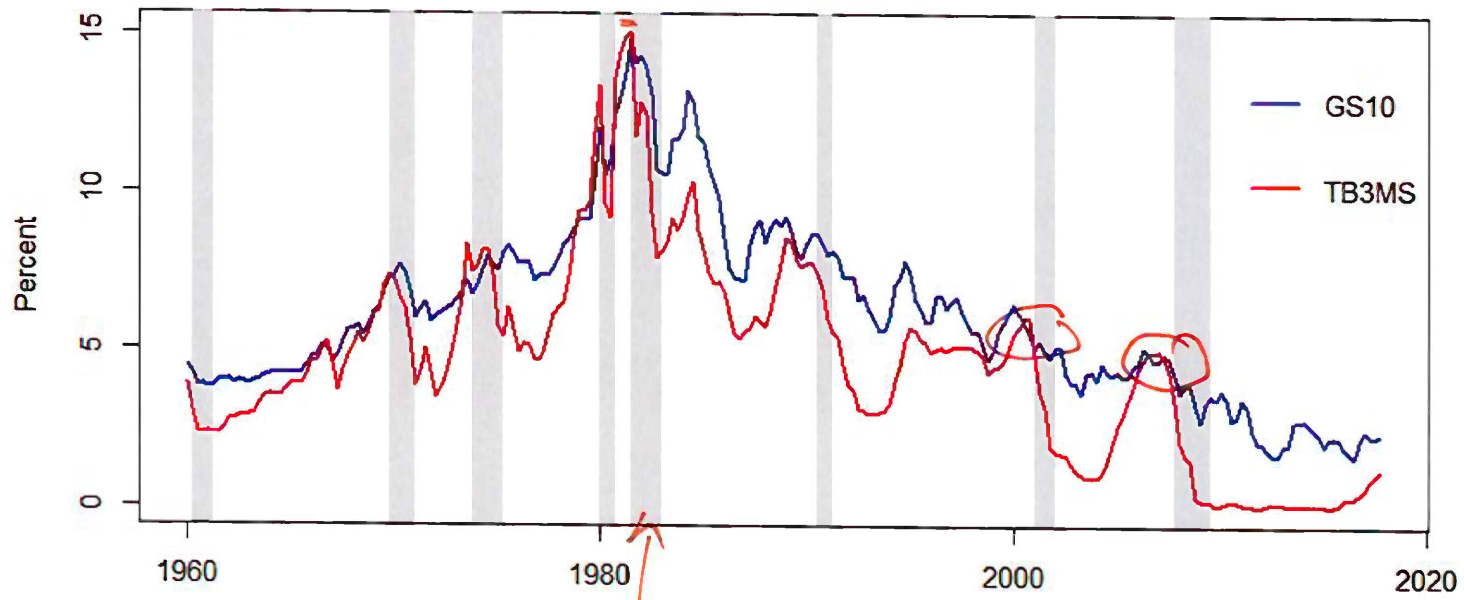
This difference between long-term and short-term interest rates is called the term spread and is plotted in the next slide.

The term spread is generally positive, but it falls toward or below 0 before recessions.

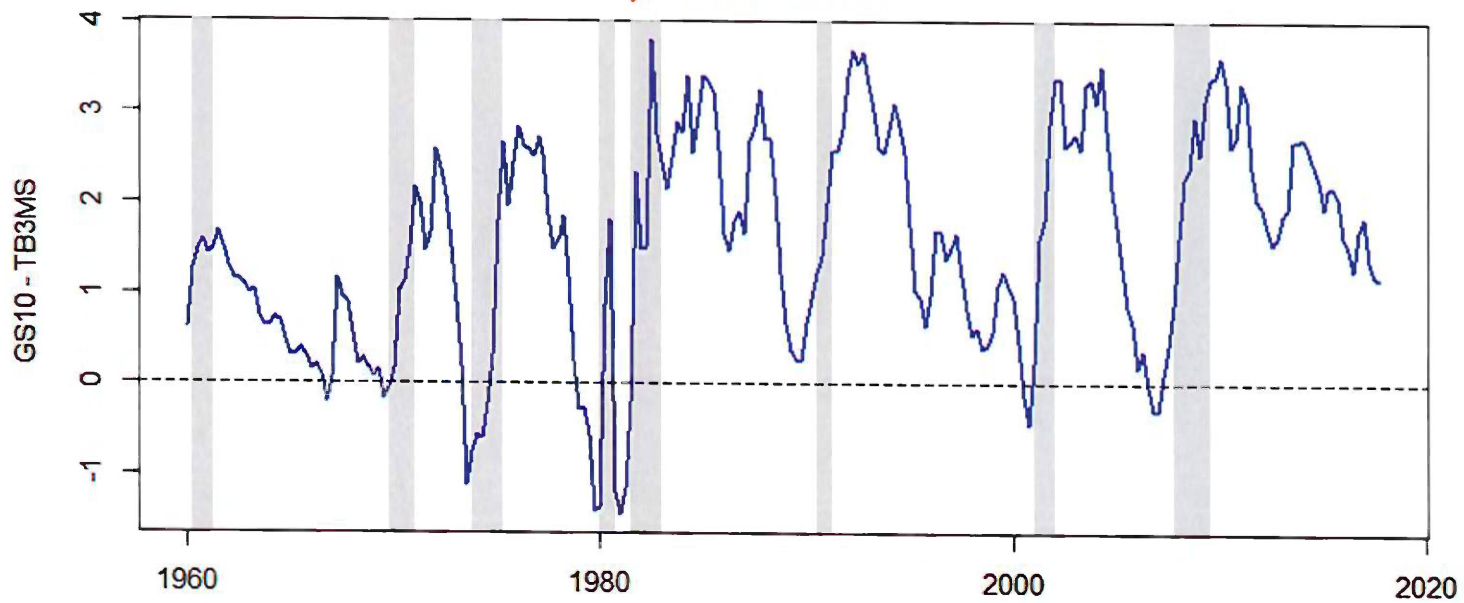


The Figure suggests that the term spread **might contain information** about the **future GDP** growth that is not already contained in past values of GDP growth.

GS10 and TB3MS



Term Spread



Construct GDP growth rate.

```
> macro.xts$GDPGR <- diff(log(macro.xts$GDP))
> tail(macro.xts)
```

annualise → from quarterly

* 400

term-spread .

growth rate

		GS10	TB3MS	RSPREAD	RECESSION	\$2009 real GDP	GDPGR
2016	Q3	1.563333	0.2966667	1.266667	0	16778.15	2.742267
2016	Q4	2.130000	0.4300000	1.700000	0	16851.42	1.743040
2017	Q1	2.443333	0.5900000	1.853333	0	16903.24	1.228157
2017	Q2	2.263333	0.8900000	1.373333	0	17031.08	3.013954
2017	Q3	2.243333	1.0366667	1.206667	0	17163.89	3.107115
2017	Q4	2.370000	1.2066667	1.163333	0	17271.70	2.504579

estimation

Final forecast ⇒

p.a. (%)

"match"

% p.a.

→ except

We use data starting in 1962:Q1 and reserve the final observation, 2017:Q4, to illustrate computing the forecast and forecast error. Thus the estimation period is from 1962:Q1–2017:Q3.

The conjecture is readily checked by augmenting the AR(2) model to include the first lag of the term spread:

$$\widehat{GDPGR}_t = 0.94 + 0.27 \overset{3.11}{GDPGR}_{t-1} + 0.19 \overset{3.01}{GDPGR}_{t-2} + 0.42 \overset{\text{slide 82}}{TSpread}_{t-1}.$$

Forecast Q4

Q3

Q2

Q3 ↑

The forecast of the rate of GDP growth in 2017:Q4 is obtained by substituting the 2017:Q2 (which is 3.01) and 2017:Q3 (which is 3.11) values of GDP growth, along with the value of the term spread in 2017:Q3 (which is 1.21); the resulting forecast is

$$\widehat{GDPGR}_{2017:Q4|2017:Q3} = 2.9\%,$$

and the forecast error is -0.4%.

```
# Create variables
GDPGR <- macro.xts$GDPGR
TSspread <- macro.xts$RSPREAD

# Create lagged variables
reg_data <- merge(
  GDPGR,
  lag(GDPGR, 1),
  lag(GDPGR, 2),
  lag(TSspread, 1)
)

# Rename columns explicitly
colnames(reg_data) <- c( "GDPGR", "GDPGR_lag1",
  "GDPGR_lag2",
  "TSspread_lag1"
)

# Remove missing values caused by differencing and lags
reg_data <- na.omit(reg_data)
```

```
# Subset from 1962 Q1 to 2017 Q3
reg_data <- reg_data[
  index(reg_data) >= as.yearqtr("1962 Q1", format = "%Y Q%q") &
  index(reg_data) <= as.yearqtr("2017 Q3", format = "%Y Q%q")
]

# Estimate the model
fit <- lm(
  GDPGR ~ GDPGR_lag1 + GDPGR_lag2 + Tspread_lag1,
  data = as.data.frame(reg_data)
)

summary(fit)
```



```
> summary(fit)
```

call:

```
lm(formula = GDPGR ~ GDPGR_lag1 + GDPGR_lag2 + TSpread_lag1,  
    data = as.data.frame(reg_data))
```

Residuals:

	Min	1Q	Median	3Q	Max
	-10.3428	-1.6306	-0.1521	1.6320	13.2549

Coefficients:

slide 83

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	0.94082	0.39733	2.368	0.01876	*
GDPGR_lag1	0.26806	0.06546	4.095	5.95e-05	***
GDPGR_lag2	0.18975	0.06534	2.904	0.00406	**
TSpread_lag1	0.42142	0.16626	2.535	0.01195	*

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.976 on 219 degrees of freedom

Multiple R-squared: 0.1697, Adjusted R-squared: 0.1583

F-statistic: 14.92 on 3 and 219 DF, p-value: 7.14e-09

Calculation

The forecast error is defined as

$$\text{Forecast error} = GDPGR_{2017:Q4} - \widehat{GDPGR}_{2017:Q4|2017:Q3}.$$

Using the given values,

$$\begin{aligned}\widehat{GDPGR}_{2017:Q4|2017:Q3} &= 0.94 + 0.27(3.11) + 0.19(3.01) + 0.42(1.21) \\ &\approx 2.9\%.\end{aligned}$$

slide 82

The actual GDP growth in 2017:Q4 is 2.5%, then the forecast error is

$$\text{Forecast error} = 2.5 - 2.8598 = -0.3598 \approx -0.4.$$

We say the forecast error is approximately

−0.4 percentage points.

↑
unit ⇒ compare % with %

Week 12: Forecasting Volatility

Consider the GARCH(1,1) model

$$\begin{aligned}r_t &= \mu + \varepsilon_t, \\ \varepsilon_t &= \sigma_t u_t, \\ \sigma_t^2 &= \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2.\end{aligned}$$

Assume

$$E(u_t \mid \mathcal{F}_{t-1}) = 0 \quad \text{and} \quad E(u_t^2 \mid \mathcal{F}_{t-1}) = 1.$$

Let $\mathcal{F}_t$ denote the information set available at time t . We want to derive the h -step-ahead forecast

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t).$$

Assume that

$$\alpha_0 > 0, \quad \alpha_1 \geq 0, \quad \beta_1 \geq 0.$$

For stationarity we require

$$\alpha_1 + \beta_1 < 1.$$

For $h \geq 2$ show that

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

Using the Law of Iterated Expectations,

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E \left[E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) \mid \mathcal{F}_t \right].$$

Since

$$\varepsilon_{t+h-1} = \sigma_{t+h-1} u_{t+h-1},$$

we have

$$\varepsilon_{t+h-1}^2 = \sigma_{t+h-1}^2 u_{t+h-1}^2.$$

Therefore, the inner expectation becomes

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) = E(\sigma_{t+h-1}^2 u_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}).$$

Given $\mathcal{F}_{t+h-2}$, the conditional variance σ_{t+h-1}^2 is known because

$$\sigma_{t+h-1}^2 = \alpha_0 + \alpha_1 \varepsilon_{t+h-2}^2 + \beta_1 \sigma_{t+h-2}^2.$$

Hence,

$$E(\sigma_{t+h-1}^2 u_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) = \sigma_{t+h-1}^2 E(u_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}).$$

Since

$$E(u_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) = 1,$$

it follows that

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) = \sigma_{t+h-1}^2.$$

Substituting this into the LIE expression,

$$\begin{aligned} E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) &= E \left[E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_{t+h-2}) \mid \mathcal{F}_t \right] \\ &= E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Thus,

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

One-Step-Ahead Forecast

For time $t + 1$, the conditional variance is

$$\sigma_{t+1}^2 = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+1}^2 \mid \mathcal{F}_t) &= E(\alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2. \end{aligned}$$

Therefore,

$$E(\sigma_{t+1}^2 \mid \mathcal{F}_t) = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2.$$

This is because ε_t^2 and σ_t^2 are known at time t .

Two-Step-Ahead Forecast

For time $t + 2$,

$$\sigma_{t+2}^2 = \alpha_0 + \alpha_1 \varepsilon_{t+1}^2 + \beta_1 \sigma_{t+1}^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+2}^2 \mid \mathcal{F}_t) &= E(\alpha_0 + \alpha_1 \varepsilon_{t+1}^2 + \beta_1 \sigma_{t+1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + \alpha_1 E(\varepsilon_{t+1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Two-Step-Ahead Forecast

Since


$$E(\varepsilon_{t+1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Substituting this into the two-step-ahead forecast,

$$\begin{aligned} E(\sigma_{t+2}^2 \mid \mathcal{F}_t) &= \alpha_0 + \alpha_1 E(\sigma_{t+1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Thus,

$$E(\sigma_{t+2}^2 \mid \mathcal{F}_t) = \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Three-Step-Ahead Forecast

For time $t + 3$,

$$\sigma_{t+3}^2 = \alpha_0 + \alpha_1 \varepsilon_{t+2}^2 + \beta_1 \sigma_{t+2}^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= E(\alpha_0 + \alpha_1 \varepsilon_{t+2}^2 + \beta_1 \sigma_{t+2}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + \alpha_1 E(\varepsilon_{t+2}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+2}^2 \mid \mathcal{F}_t). \end{aligned}$$

$$\downarrow \\ E(\varepsilon_{t+2}^2 \mid \mathcal{F}_t) \Rightarrow \text{slide 4}$$

Three-Step-Ahead Forecast

Since

$$E(\varepsilon_{t+2}^2 \mid \mathcal{F}_t) = E(\sigma_{t+2}^2 \mid \mathcal{F}_t),$$

we obtain

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= \alpha_0 + \alpha_1 \overbrace{E(\sigma_{t+2}^2 \mid \mathcal{F}_t)} + \beta_1 \overbrace{E(\sigma_{t+2}^2 \mid \mathcal{F}_t)} \\ &= \alpha_0 + (\alpha_1 + \beta_1) \underbrace{E(\sigma_{t+2}^2 \mid \mathcal{F}_t)}_{\uparrow}. \end{aligned}$$

Therefore,

$$E(\sigma_{t+3}^2 \mid \mathcal{F}_t) = \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+2}^2 \mid \mathcal{F}_t).$$

Three-Step-Ahead Forecast

Using the two-step-ahead forecast,

$$E(\sigma_{t+2}^2 \mid \mathcal{F}_t) = \alpha_0 + \underbrace{(\alpha_1 + \beta_1)E(\sigma_{t+1}^2 \mid \mathcal{F}_t)},$$

we get

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= \alpha_0 + (\alpha_1 + \beta_1) [\alpha_0 + (\alpha_1 + \beta_1)E(\sigma_{t+1}^2 \mid \mathcal{F}_t)] \\ &= \alpha_0 + \alpha_0(\alpha_1 + \beta_1) + (\alpha_1 + \beta_1)^2 E(\sigma_{t+1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 [1 + (\alpha_1 + \beta_1)] + (\alpha_1 + \beta_1)^2 E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

General h -Step-Ahead Forecast

For $h \geq 2$, the GARCH equation is

$$\sigma_{t+h}^2 = \alpha_0 + \alpha_1 \varepsilon_{t+h-1}^2 + \beta_1 \sigma_{t+h-1}^2.$$

Taking conditional expectations given $\mathcal{F}_t$,

$$\begin{aligned} E(\sigma_{t+h}^2 \mid \mathcal{F}_t) &= E(\alpha_0 + \alpha_1 \varepsilon_{t+h-1}^2 + \beta_1 \sigma_{t+h-1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + \alpha_1 E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t). \end{aligned}$$

$$\underbrace{E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t)}_{\text{slide 4}}$$

General h -Step-Ahead Forecast

Since

$$E(\varepsilon_{t+h-1}^2 \mid \mathcal{F}_t) = E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

Substituting this result into the forecast equation gives

$$\begin{aligned} E(\sigma_{t+h}^2 \mid \mathcal{F}_t) &= \alpha_0 + \alpha_1 E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t) + \beta_1 E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t) \\ &= \alpha_0 + (\alpha_1 + \beta_1) \underbrace{E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t)}. \end{aligned}$$

Therefore, for $h \geq 2$,

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \alpha_0 + (\alpha_1 + \beta_1) E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

Closed-Form Solution

Let

$$\rho = \alpha_1 + \beta_1. \quad < \mid$$

Then, for $h \geq 2$, the recursive forecast is

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \alpha_0 + \rho E(\sigma_{t+h-1}^2 \mid \mathcal{F}_t).$$

Starting from the one-step-ahead forecast,

$$E(\sigma_{t+1}^2 \mid \mathcal{F}_t) = \alpha_0 + \alpha_1 \varepsilon_t^2 + \beta_1 \sigma_t^2,$$

we obtain

$$E(\sigma_{t+2}^2 \mid \mathcal{F}_t) = \alpha_0 + \rho E(\sigma_{t+1}^2 \mid \mathcal{F}_t),$$

and

$$E(\sigma_{t+3}^2 \mid \mathcal{F}_t) = \alpha_0 + \rho E(\sigma_{t+2}^2 \mid \mathcal{F}_t).$$

Substituting recursively,

$$\begin{aligned} E(\sigma_{t+3}^2 \mid \mathcal{F}_t) &= \alpha_0 + \rho [\alpha_0 + \rho E(\sigma_{t+1}^2 \mid \mathcal{F}_t)] \\ &= \alpha_0(1 + \rho) + \rho^2 E(\sigma_{t+1}^2 \mid \mathcal{F}_t). \end{aligned}$$

Continuing this pattern,

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \alpha_0 \underbrace{(1 + \rho + \rho^2 + \cdots + \rho^{h-2})}_{\text{red underline}} + \rho^{h-1} E(\sigma_{t+1}^2 \mid \mathcal{F}_t).$$

Geometric Series

Using the geometric series,

$$1 + \rho + \rho^2 + \cdots + \rho^{h-2} = \frac{1 - \rho^{h-1}}{1 - \rho},$$

General h -Step-Ahead Forecast

$$h \rightarrow \infty \quad \rho < 1 \\ \hookrightarrow \rho^{h-1} \approx 0$$

We now have

$$E(\sigma_{t+h}^2 | \mathcal{F}_t) = \alpha_0 \frac{1 - \rho^{h-1}}{1 - \rho} + \rho^{h-1} E(\sigma_{t+1}^2 | \mathcal{F}_t).$$

Since $\rho = \alpha_1 + \beta_1$, this can be written as

$$E(\sigma_{t+h}^2 | \mathcal{F}_t) = \alpha_0 \frac{1 - (\alpha_1 + \beta_1)^{h-1}}{1 - (\alpha_1 + \beta_1)} + (\alpha_1 + \beta_1)^{h-1} E(\sigma_{t+1}^2 | \mathcal{F}_t).$$

as $h \rightarrow \infty \Rightarrow \frac{\alpha_0}{1 - \rho} \Rightarrow \frac{\alpha_0}{1 - (\alpha_1 + \beta_1)} \Rightarrow \text{LR variance.}$

Recall that the unconditional variance is

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \alpha_1 - \beta_1}. \quad = LR$$

Since $\rho = \alpha_1 + \beta_1$, we have

$$\bar{\sigma}^2 = \frac{\alpha_0}{1 - \rho}.$$

Therefore, the h -step-ahead forecast can be written as

$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) = \bar{\sigma}^2 + (\alpha_1 + \beta_1)^{h-1} [E(\sigma_{t+1}^2 \mid \mathcal{F}_t) - \bar{\sigma}^2].$$



$h \rightarrow \infty$

becomes zero

As $h \rightarrow \infty$,

$$(\alpha_1 + \beta_1)^{h-1} \rightarrow 0,$$

provided that

$$\alpha_1 + \beta_1 < 1.$$

Hence,

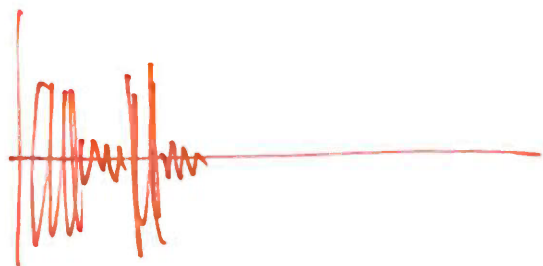
$$E(\sigma_{t+h}^2 \mid \mathcal{F}_t) \rightarrow \bar{\sigma}^2 \quad \text{as } h \rightarrow \infty.$$

Therefore, GARCH variance forecasts mean-revert to the unconditional variance.

Stylised Facts

The ARCH(1) and GARCH(1,1) models capture several **stylised facts** of asset returns:

- ▶ **Volatility clustering:** large shocks tend to be followed by large shocks, and small shocks tend to be followed by small shocks.
- ▶ **Time-varying volatility:** the conditional variance changes over time depending on past shocks and/or past volatility.



Stylised Fact Not Captured by ARCH(1) and GARCH(1,1)

The standard ARCH(1) and GARCH(1,1) models do not capture the **leverage effect**, because positive and negative shocks of the same size have the same effect on future volatility.

For leverage effects, we usually need models such as GJR-GARCH or EGARCH.

Revision

MDS

Definition

A time series ε_t is a martingale difference sequence (MDS) if

$$E(\varepsilon_t \mid \mathcal{F}_{t-1}) = 0$$

Here $\mathcal{F}_{t-1}$ denotes the information set available up to time $t - 1$.

workshop week 8 $\Rightarrow E(\varepsilon_t) = 0$

$$\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = 0 \quad k \geq 1$$

White Noise

A time series $\{\epsilon_t\}_{t \geq 1}$ is called a **white noise** process if

1. $\mathbb{E}[\epsilon_t] = 0$;
2. $\mathbb{V}[\epsilon_t] = \sigma_\epsilon^2$ for all t ;
3. $\gamma_\epsilon(k) = \text{Corr}(\epsilon_t, \epsilon_{t-k}) = 0$, for all t and k .

This is sometimes written as $\epsilon_t \sim \text{WN}(0, \sigma_\epsilon^2)$.

independently and identically distributed (i.i.d.)

u_t is assumed to be independently and identically distributed for all t (i.i.d.), that is

$$u_t \sim \text{i.i.d. } (0, \sigma^2).$$

This notation means that u_t has mean zero and variance σ^2 .

The assumption that u_t is independent over time means that it has no autocorrelation at any lag.

↗ any linear $\Leftrightarrow$ non-linear

$\text{Cov}(u_t, u_{t-k}) \Rightarrow$ linear relationship.

It also has a constant variance σ^2 over time so it is homoskedastic.

A time series that has no autocorrelation and is homoskedastic is sometimes called *white noise*. Therefore u_t is an example of a white noise process.

Stationary

A time series Y_t is defined to be covariance *stationary* if

1. $E(Y_t) = \mu$ does not depend on t
2. $\text{var}(Y_t) = \sigma^2$ does not depend on t
3. $\text{cov}(Y_t, Y_{t-j}) = \gamma_j$ depends only on j but not on t , for $j = 1, 2, \dots$
 $\neq 0$